Problem Statement

Let $X \in \mathbb{R}^{n \times p_1}$ denote a matrix of activations of p_1 neurons for n examples, and $Y \in \mathbb{R}^{n \times p_2}$ denote a matrix of activations of p_2 neurons for the same n examples. We assume that these matrices have been preprocessed to center the columns. Without loss of generality we assume that $p_1 \leq p_2$. We are concerned with the design and analysis of a scalar *similarity index* $s(X, Y)$ that can be used to compare representations within and across neural networks, in order to help visualize and understand the effect of different factors of variation in deep learning.

2. What Should Similarity Be Invariant To?

This section discusses the invariance properties of similarity indexes and their implications for measuring similarity of neural network representations. We argue that both intuitive notions of similarity and the dynamics of neural network training call for a similarity index that is invariant to orthogonal transformation and isotropic scaling, but not invertible linear transformation.

2.1. Invariance to Invertible Linear Transformation

A similarity index is invariant to invertible linear transformation if $s(X, Y) = s(XA, YB)$ for any full rank A and B. If activations X are followed by a fully-connected layer $f(X) = \sigma(XW + \beta)$, then transforming the activations by a full rank matrix A as $X' = XA$ and transforming the weights by the inverse A^{-1} as $W' = A^{-1}W$ preserves the output of $f(X)$. This transformation does not appear to change how the network operates, so intuitively, one might prefer a similarity index that is invariant to invertible linear transformation, as argued by Raghu et al. (2017).

However, a limitation of invariance to invertible linear transformation is that any invariant similarity index gives the same result for any representation of width greater than or equal to the dataset size, *i.e.* $p_2 \geq n$. We provide a simple proof in Appendix A.

Theorem 1. *Let X and Y be $n \times p$ matrices. Suppose s is invariant to invertible linear transformation in the first argument,* i.e. $s(X, Z) = s(XA, Z)$ *for arbitrary Z and any A with rank$(A) = p$. If rank(X) = rank$(Y) = n$, then $s(X, Z) = s(Y, Z)$.*

There is thus a practical problem with invariance to invertible linear transformation: Some neural networks, especially convolutional networks, have more neurons in some layers than there are examples the training dataset (Springenberg et al., 2015; Lee et al., 2018; Zagoruyko & Komodakis, 2016). It is somewhat unnatural that a similarity index could require more examples than were used for training.

A deeper issue is that neural network *training* is not invariant to arbitrary invertible linear transformation of inputs or activations. Even in the linear case, gradient descent converges first along the eigenvectors corresponding to the largest eigenvalues of the input covariance matrix (LeCun et al., 1991), and in cases of overparameterization or early stopping, the solution reached depends on the scale of the input. Similar results hold for gradient descent training of neural networks in the infinite width limit (Jacot et al., 2018). The sensitivity of neural networks training to linear transformation is further demonstrated by the popularity of batch normalization (Ioffe & Szegedy, 2015).

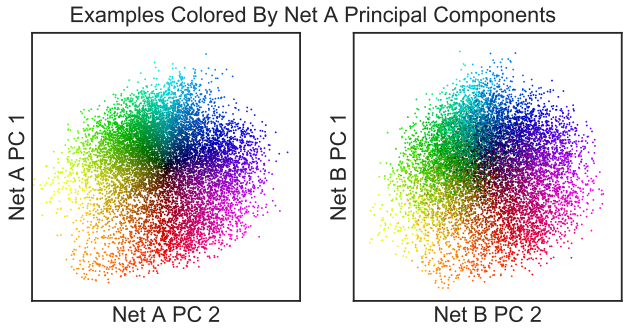

Figure 1. First principal components of representations of networks trained from different random initializations are similar. Each example from the CIFAR-10 test set is shown as a dot colored according to the value of the first two principal components of an intermediate layer of one network (left) and plotted on the first two principal components of the same layer of an architecturally identical network trained from a different initialization (right).

Invariance to invertible linear transformation implies that the scale of directions in activation space is irrelevant. Empirically, however, scale information is both consistent across networks and useful across tasks. Neural networks trained from different random initializations develop representations with similar large principal components, as shown in Figure 1. Consequently, Euclidean distances between examples, which depend primarily upon large principal components, are similar across networks. These distances are *meaningful*, as demonstrated by the success of perceptual loss and style transfer (Gatys et al., 2016; Johnson et al., 2016; Dumoulin et al., 2017). A similarity index that is invariant to invertible linear transformation ignores this aspect of the representation, and assigns the same score to networks that match only in large principal components or networks that match only in small principal components.

2.2. Invariance to Orthogonal Transformation

Rather than requiring invariance to any invertible linear transformation, one could require a weaker condition; invariance to orthogonal transformation, *i.e.* $s(X, Y) = s(XU, YV)$ for full-rank orthonormal matrices U and V such that $U^{\mathrm{T}}U = I$ and $V^{\mathrm{T}}V = I$.